Assignment #2 – Implementation of cryptography in planned scenario

OBJECTIVE:

1. Implement the cryptographic mechanisms to secure data at transit in services.
2. Implement the cryptographic mechanisms to secure data at rest in services.
3. Validate some components/mechanisms in the CBOM – Cryptographic Bill of Materials.

CONTEXT:

You are responsible to implement the scenario that you have planned in the previous assignment. Main the minimal set of services (three services, where the DB is already included). Please consider the assignment 1 regarding the introduction of the services and CBOM.

Assignment:

The assignment includes different tasks, including the implementation of the choices made in the first assignment. In particular **at least two services + Database** are to be instantiated according to the plan[1].

1. IMPLEMENT THE SERVICES

The first task is to implement the services according to the plan. If you have decided to use containers, virtual machines or other. You need to deploy the services and to configure them in a secure way.

You need to:

a. Implement the services and document the main configurations.
b. Document the steps followed to implement the data flows (identified in the planning) and the basic functionalities like authentication, authorization.
c. Present a diagram with the overall instantiation of the services, highlighting the used ports, protocols, if deployed as containers and/or virtual machines.
d. Document also the policies you have regarding the maintenance of key material, and the security category. For instance, what is the renewal of keys/certificates.

[1] The plan in the assignment 1 can be reformulated, during the execution of the assignment 2. In such case the report of the second assignment should document the changes, motivations, technical barriers, issues among other aspects

2. VALIDATE SOME COMPONENTS OF THE CBOM

In this task you need to document some of components of the CBOM that you have identified in the planning and that you have confirmed/configured in the second assignment.
For instance, if you solution uses TLS, which algorithms have you used to configure the encryption, the integrity control, among other aspects.

You need to:

a. Document the components you have configured and that were/are part of the CBOM. Explain the main reasons to document such components and justify the usage of some algorithms and keys sizes. The justification can be based on the NIST security categories document in NIST SP 800-57.

3. MANDATORY POLICIES

In the implementation the following policies must be implemented, according to the services you have selected.

One of the key services in the second assignment is a PKI that is important for certificate management. Some of the services need to support certificates.

Table 1 – Mandatory policies for services.

Service	Policies
Frontend	• Code/configuration integrity control, assure you have mechanisms to protect the code or the configurations of the service. • Support encryption of data in transit. • Support the use of X509 certificates.
Backend	• Support encryption of data in transit. • Integrity control of code/configuration files. • Support the use of X509 certificates.
Database	• Encrypt data at rest. • Encrypt data at transit. • Integrity control of configuration files • Authentication, accounting and authorization of users/devices/services • Support the use of X509 certificates.
Third-Party	• Support integrity control of configuration files, running code. • Configure robust hashing and encryption mechanisms. • (as PKI) support management of X509 certificates (issuance, revoking). • (as PKI) support status verification of X509 certificates either with CRLs and/or using OCSP.

NOTE:
Some of these policies might not have been planned initially in the first assignment. Consider in reviewing the planning and document the necessary changes to support these policies.

REPORT (WHAT TO DELIVER):

The report should document in detail the implementation that was done, answering all the aspects asked.

Plan your work to be delivered on 2025-12-15.